

$\sqrt{g}$ - **vol** ; h_x, h_y, h_z - **hx,hy,hz** ; B_x, B_z - **bbx,bbz** ; B - **bb**

crx0, cry0 physical Cartesian coordinates (y axis coincides with the symmetry axis of tokamak) of the left lower corner of the cell in the mesh transformed to the orthogonal one.

$$\frac{n_a}{t} \frac{1}{\sqrt{g}} \frac{\sqrt{g}}{h_x} \sim_{ax} \frac{1}{\sqrt{g}} \frac{\sqrt{g}}{h_y} \sim_{ay} S_a^n, a = 0, 1, \dots, n_s-1 \quad (1)$$

The residuals of the iteration of this equation are **b2npc9_resco01** for ions and **b2npco_resco00** for neutrals.

n_a - **b2npc9_na01** for ions and **b2npco_na00** for neutrals

$S_a^n \sqrt{g}$ - **b2npc9_sna01** for ions and **b2npco_sna00** for neutrals

$S_a^n \sqrt{g} - S_a^{ion} \sqrt{g} - S_a^{rec} \sqrt{g}$ These sources are in **b2stel_sna_ion** and **b2stel_sna_rec** respectively.

$\frac{n_a}{t} \sqrt{g}$ - **b2srdt_snadt01** for ions and **b2srdt_snadt00** for neutrals

$\frac{\sqrt{g}}{h_x} \sim_{ax}$ - **b2npc9_fnax01** for ions and **b2npco_fnax00** for neutrals

$\frac{\sqrt{g}}{h_x} \sim_{ay}$ - **b2npc9_fnay01** for ions and **b2npco_fnay00** for neutrals

$$\sim_{ax} b_x V_{\parallel a} n_a V_x^{(E)} n_a V_{ax}^{(AN)} n_a V^{corr-dpc} n_a D_a^n \frac{1}{h_x} \frac{n_a}{x} \tilde{D}_{ax}^p \frac{1}{h_x} \frac{p_a}{x} D_{axy} \frac{1}{h_y} \frac{n_a T_i}{y}$$

$$\sim_{ay} V_y^{(E)} n_a V_{ay}^{(AN)} n_a \frac{j_y^{(AN)} j_y^{(in)} \tilde{j}_y^{(vis)}}{e} n_a D_a^n \frac{1}{h_y} \frac{n_a}{y} \tilde{D}_{ay}^p \frac{1}{h_y} \frac{p_a}{y} D_{ayx} \frac{1}{h_x} \frac{n_a T_i}{x},$$

$\frac{\sqrt{g}}{h_x} b_x V_{\parallel a} n_a$ - **b2tfnb_bxuanax01** for ions and **b2tfnb_bxuanax00** for neutrals

$\frac{\sqrt{g}}{h_x} V_{x,y}^{(E)} n_a$ - **b2tfnb_vaecrbnax,y01** for ions and **0** for neutrals

$\frac{\sqrt{g}}{h_x} V^{corr-dpc} n_a$ - **b2tfnb_dpccornax01** for ions and **b2tfnb_dpccornax00** for neutrals

$\frac{\sqrt{g}}{h_x} V_{ay}^{(AN)} n_a$ - **b2tfnb_cvlbnay01** for ions

$$D_a^n \frac{\sqrt{g}}{h_x} \frac{1}{h_x} \frac{n_a}{x}, D_a^n \frac{\sqrt{g}}{h_y} \frac{1}{h_y} \frac{n_a}{y} - \mathbf{b2tfnb_dPat_mdf_gradna\{x,y\}01} \text{ for ions and}$$

b2tfnb_dPat_mdf_gradnax,y00 for neutrals. These are the values which actually enter the particle balance in the code. Diffusion coefficients are modified for the numerical stability according to Potanko scheme :

$$D_{x,y}^{modified} = \sqrt{D_{x,y}^2 - (V_{x,y}^{convective} h_{x,y} / 2)^2}. \text{ For neutrals these fluxes are usually } 0.$$

$D_a^n \frac{\sqrt{g}}{h_{x,y}^2} - \mathbf{b2tral_cdnax,y}$ diffusion coefficients before the Potanco correction.

$\tilde{D}_{ax}^p \frac{\sqrt{g}}{h_x} \frac{1}{h_x} \frac{p_a}{x}$, $\tilde{D}_{ax}^p \frac{\sqrt{g}}{h_x} \frac{1}{h_x} \frac{p_a}{x} - \mathbf{b2tfnb_dgradpbx,y01}$ for ions and $\mathbf{b2tfnb_dgradpbx,y00}$ for neutrals. For ions these fluxes are usually 0.

$\tilde{D}_{ax,y}^p \frac{\sqrt{g}}{h_{x,y}^2} - \mathbf{b2tral_cdpax,y}$

$D_{axy} \frac{\sqrt{g}}{h_x} \frac{1}{h_y} \frac{n_a T_i}{y}$, $D_{ayx} \frac{\sqrt{g}}{h_y} \frac{1}{h_x} \frac{n_a T_i}{x} - \mathbf{b2tfnb_fnbPSchx,y01}$ for ions and 0 for neutrals. Mind the sign! It is opposite in output file and in flux balance.

$D_{axy,yx} \frac{\sqrt{g}}{h_{x,y}^2} - \mathbf{b2tral_cddix,y}$

$V^{corr-dpc} = \frac{c^{corr-dpc}}{16} \frac{B_x}{B} \sqrt{\frac{p_a}{m_a n_a} \frac{1}{x^2}} \frac{1}{p_a} \frac{p_a}{x}$

$V_x^{(E)} = c_{EB} \frac{B_z}{B^2} \frac{1}{h_y} \frac{1}{y}$, $z_a = 0$, $V_y^{(E)} = c_{EB} \frac{B_z}{B^2} \frac{1}{h_x} \frac{1}{x}$, $z_a = 0$,
0, $z_a = 0$ 0, $z_a = 0$

$\tilde{V}_{xa}^{(dia)} = c_{dia} \frac{T_i B_z}{Z_a e h_y} \frac{1}{y} \frac{1}{B^2}$, $z_a = 0$, $\tilde{V}_{ya}^{(dia)} = c_{dia} \frac{T_i B_z}{Z_a e h_x} \frac{1}{x} \frac{1}{B^2}$, $z_a = 0$,
0, $z_a = 0$ 0, $z_a = 0$

$\frac{\sqrt{g}}{h_x} \tilde{V}_{x,y1}^{(dia)} n_a - \mathbf{b2tfnb_vadianax,y01}$ for ions and 0 for neutrals

$D_{axy} = \frac{1}{B^2} \frac{1}{\langle B^2 \rangle} \frac{B_z}{z_a e}$, $z_a = 0$, where $\langle B^2 \rangle = \frac{B^2 \sqrt{g} dx dy}{\int \sqrt{g} dx dy}$
 $D_{axy} = 0$, $z_a = 0$ core

$D_{ayx} = \frac{1}{B^2} \frac{1}{\langle B^2 \rangle} \frac{B_z}{z_a e}$, $z_a = 0$
 $D_{ayx} = 0$, $z_a = 0$

$n_{e,a-1} z_a n_a$, $n_{i,a-1} n_a$, $p_a = n_a z_a T_e$ T_i

$D_a^n = D_{ANa}^n$ $D_{ANa}^n = c_{D_a^N,0} c_{D_a^N,1} \frac{10^{20}}{n_e} c_{D_a^N,2} D_a^f c_{D_a^N,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} f_D(x,y)$

$$f_D(x, y) = \begin{cases} 1, & z_a = 0 \\ 1, D, & y = 0, x \in Core, z_a = 0 \\ 2, D, & 0 < y < 2, D, x \in Core, \\ 3, D, & x, y \in Pr \text{ SOL} \end{cases}$$

$$a = 0, c_{D_a^N, 0} = 0, c_{D_a^N, 1} = 0, c_{D_a^N, 2} = 0, c_{D_a^N, 3} = 0$$

$$a = 1, c_{D_a^N, 0} = 0, c_{D_a^N, 1} = 0, c_{D_a^N, 2} = 1, c_{D_a^N, 3} = 0$$

$$D_a^f = c_{D_a^f, 0} + c_{D_a^f, 1} \frac{10^{-20}}{n_i} + \frac{c_{D_a^f, 2} \sqrt{\frac{T_i}{m_a}}}{c_{D_a^f, 3} n_i + c_{D_a^f, 4} n_e},$$

a	$c_{D_a^f, 0}$	$c_{D_a^f, 1}$	$c_{D_a^f, 2}$	$c_{D_a^f, 3}$	$c_{D_a^f, 4}$	$c_{D_a^f, 5}$	$c_{D_a^f, 6}$	$c_{D_a^f, 7}$
0	10^{-2}	0	1	$5 \cdot 10^{-20}$	$1,5 \cdot 10^{-18}$	0	0	0
1	0,3	0	0	—	—	0	0	0

$$\tilde{D}_{ax}^p = 1 - \frac{D_a^p \left| \frac{1}{h_x} - \frac{p_a}{x} \right|}{n_a \frac{V_{T_i}}{4}} \sqrt{\frac{8T_i}{m_a}}, \quad z_a = 0 \quad \tilde{D}_{ay}^p = 1 - \frac{D_a^p \left| \frac{1}{h_y} - \frac{p_a}{y} \right|}{n_a \frac{V_{T_i}}{4}} \sqrt{\frac{8T_i}{m_a}}, \quad z_a = 0$$

$$D_a^p, \quad z_a = 0 \quad D_a^p, \quad z_a = 0$$

$$D_a^p = D_{ANa}^p$$

$$D_{ANa}^p = \frac{1}{z_a T_e T_i} c_{D_a^p, 0} + c_{D_a^p, 1} \frac{10^{-20}}{n_e} + c_{D_a^p, 2} D_a^f + c_{D_a^p, 3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|},$$

$$a = 0, c_{D_a^p, 0} = 0, c_{D_a^p, 1} = 0, c_{D_a^p, 2} = 1, c_{D_a^p, 3} = 0$$

$$a = 1, c_{D_a^p, 0} = 0, c_{D_a^p, 1} = 0, c_{D_a^p, 2} = 0, c_{D_a^p, 3} = 0$$

$$V_{ax}^{(AN)} = 0$$

$$V_{ay}^{(AN)} = c_{V_a^{(AN)}, 0} + c_{V_a^{(AN)}, 1} \frac{10^{-20}}{n_e} + c_{V_a^{(AN)}, 2} D_a^f + c_{V_a^{(AN)}, 3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|}$$

$$a = 0, c_{V_a^{(AN)}, 0} = 0, c_{V_a^{(AN)}, 1} = 0, c_{V_a^{(AN)}, 2} = 0, c_{V_a^{(AN)}, 3} = 0$$

$$a = 1, c_{V_a^{(AN)}, 0} = 0, c_{V_a^{(AN)}, 1} = 0, c_{V_a^{(AN)}, 2} = 0, c_{V_a^{(AN)}, 3} = 0$$

$$m_a \frac{n_a V_{\parallel a}}{t} - \frac{1}{h_z \sqrt{g}} \frac{h_z \sqrt{g}}{h_x} \frac{m}{ax} - \frac{1}{h_z \sqrt{g}} \frac{h_z \sqrt{g}}{h_y} \frac{m}{ay} - \frac{b_x}{h_x} \frac{n_a T_i}{x} - \frac{b_x}{h_x} \frac{Z_a n_a n_e T_e}{n_e x} S_{a\parallel}^m S_{CF}^m S_{fr}^m \quad (2)$$

$V_{\parallel a}$ - **b2npmo_ua** Index **01** is for ions and **00** for neutrals everywhere afterwards

The residuals of the iteration of this equation are **b2npmo_resmo01** for ions and **b2npmo_resmo00** for neutrals.

$S_a^m h_z \sqrt{g}$ - **b2npmo_smb** : full r.h.s. of the equation including pressure gradient contribution

$$m_a \frac{n_a V_{\parallel a}}{t} h_z \sqrt{g} - \mathbf{b2srdt_smodt}$$

$$\frac{h_z \sqrt{g}}{h_x} \frac{m}{ax,y} - \mathbf{b2nmo_fmax,y}$$

$$S_{CF}^m h_z \sqrt{g} - m_a b_x n_a V_{\parallel a}^2 \frac{1}{h_z h_x} \frac{h_z}{x} h_z \sqrt{g} - \mathbf{b2npmo_smocf}$$

$$\frac{b_x}{h_x} \frac{n_a T_i}{x} - \frac{b_x}{h_x} \frac{Z_a n_a n_e T_e}{n_e x} h_z \sqrt{g} - \mathbf{b2npmo_smogp}$$

$S_{fr}^m h_z \sqrt{g}$ - the sum of contributions from ion-neutral charge-exchange friction (**b2stcx_smq**), ionization (**b2stel_smq_ion**), recombination (**b2stel_smq_rec**), friction interaction between different ion species (**b2npmo_smoch**) and thermal force (**b2npmo_smotf**, zero while single ion species).

$$0, \quad a = 1,$$

$$S_{a\parallel}^m \frac{h_z \sqrt{g}}{h_x} \frac{4}{3} \frac{ax}{B^2 h_x} \frac{\ln h_z B^{\frac{1}{2}}}{B^2 h_x} B^{\frac{1}{2}} V_{\parallel a} B^{\frac{3}{2}} b_x \frac{NEO}{h_x} \frac{b_x \tilde{a}}{B^2} \frac{B^{\frac{1}{2}} \tilde{q}^{(0)}}{h_x}, \quad a = 1$$

First and second terms of this viscosity momentum source are **b2npmo_smovv** and **b2npmo_smovh** respectively.

$$m_a V_{\parallel a} \frac{ax}{h_x} - m_a V_{\parallel a} \frac{(dia)}{ax} \frac{V_{\parallel a}}{h_x}, \quad a = 1$$

$$m_a V_{\parallel a} \frac{ax}{h_x} - m_a V_{\parallel a} \frac{(dia)}{ax} \frac{4}{3} \frac{\ln h_z}{h_x} V_{\parallel a} - \frac{4}{3} \frac{V_{\parallel a}}{h_x}, \quad a = 1$$

here the contribution from classic parallel viscosity included

$$ax \frac{(dia)}{ax} \frac{h_z \sqrt{g}}{h_x} - b_x V_{\parallel a} n_a V_x^{(E)} n_a - 2 \tilde{V}_x^{(dia)} n_a V_{ax}^{(AN)} n_a - D_a^n \frac{1}{h_x} \frac{n_a}{x} - \tilde{D}_{ax}^p \frac{1}{h_x} \frac{p_a}{x} - \frac{h_z \sqrt{g}}{h_x} -$$

b2tfnb_fnb_fcorx

$$D_a^n \frac{1}{h_x} \frac{n_a}{x} \frac{h_z \sqrt{g}}{h_x} - \mathbf{b2tfnb_dPat_2diagradnax}$$
 here diffusion coefficient is modified according to

convective flux with $2\tilde{V}_x^{(dia)}$

$$\frac{h_z \sqrt{g}}{h_x} \frac{4}{3} \frac{\ln h_z}{h_x x} - \mathbf{b2nprmo_flubvx01}$$
 convective flow addition due to parallel viscosity

$$m_a V_{\parallel a} \frac{V_{\parallel a}^{(dia)}}{h_x x},$$

$$\frac{h_z \sqrt{g}}{h_x} \frac{4}{3} \frac{V_{\parallel a}}{h_x x} \text{ for } a = 1 \text{ and } \frac{V_{\parallel a}}{h_x x} \text{ for } a = 1 - \mathbf{b2urmo_etaPat_graduax01}$$

the Potanko correction is applied for ax

$$m_{ay} V_{\parallel a} \frac{V_{\parallel a}^{(dia)}}{h_y y},$$

$$\sqrt{\quad}$$

- $\mathbf{b2tfnb_fmb_fcory}$

$$m_{ay} \frac{V_{\parallel a}}{h_y y} - \mathbf{b2urmo_etaPat_graduay01}$$
 the Potanko correction is applied for ay

$$ax \quad ANa \quad \approx \quad CLax \quad \frac{h_z \sqrt{g}}{h_x^2} - \mathbf{b2tral_cvsahzx}$$

$$c_{a,0} \quad c_{a,1} \frac{10^{20}}{n_e} \quad c_{a,2} D_a^f \quad c_{a,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} m_a n_a f_a(x, y), \quad z_a = 0$$

$$ANa \quad c_{a,0} \quad c_{a,1} \frac{10^{20}}{n_e} \quad c_{a,2} D_a^f \quad c_{a,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} m_a n_a, \quad z_a = 0$$

$$1, \quad z_a = 0$$

$$f_a(x, y) \quad \begin{matrix} 1, a, & 1, a & y = 0, & x \text{ Core}, & z_a = 0 \\ 2, a, & 0 & y & 2, a, & x \text{ Core}, \\ 3, a, & x, y & \text{Pr SOL} \end{matrix}$$

$$a = 0, \quad c_{a,0} = 0, \quad c_{a,1} = 0, \quad c_{a,2} = 1, \quad c_{a,3} = 0$$

$$a = 1, \quad c_{a,0} = 0, \quad c_{a,1} = 0, \quad c_{a,2} = 1, \quad c_{a,3} = 0$$

$$\approx \quad CLax \quad C_{ax}^{(CL)} \approx \quad CLax \quad \frac{h_z \sqrt{g}}{h_x^2} - \mathbf{b2trcl_luciani_flim_cvsahzx}$$

1, $z_a = 0$ and only for inner flux surfaces

$$c_{ax}^{(CL)} = \frac{1}{1 + \frac{\sqrt{g}}{h_x} \sim_{CLax} \frac{V_{\parallel a}}{h_x x}}, \quad \text{for all others cases}$$

$$V_{m_a}^{(max)} = c_{f \lim 2} \frac{\sqrt{g}}{h_x} b_x n_a T_i$$

$$\sim_{CLax} c_{ax}^{(Luciani)} \sim_{CLax} b_x^2 \sim_{CLax} \frac{h_z \sqrt{g}}{h_x^2} - \text{b2trcl_luciani_cvsahzx}$$

For plato and banana correction ‘b2trcl_lluciani’ ‘3’

$$c_{ax}^{(Luciani)} = \frac{1}{1 + \frac{15,12 \cdot 10^{16} \cdot \frac{T_i}{e}}{c(y) \cdot z_a^2 n_a}}, \quad z_a = 0 \text{ and only for inner flux surfaces ,}$$

1, for all others cases

$$c_y = \frac{\oint h_x b_x^{-1} dx}{\oint \sqrt{g} / h_y dx}, \quad \frac{\oint h_x dx}{\oint \sqrt{g} / h_y dx}$$

$$\begin{aligned} &0, \quad z_a = 0 \\ &_{ax}^{(CL)} T_i n_a \frac{a}{\sqrt{2}}, \quad z_a = 0, \quad 1,357 \quad \frac{_{ax}^{(CL)} h_z \sqrt{g}}{h_x^2} b_x^2 - \text{b2trcl_cvsahzx} \end{aligned}$$

$$a = \frac{3}{4\sqrt{2}} \frac{\sqrt{m_a} T_i^{\frac{3}{2}}}{z_a^2 z_{eff} n_e} - \frac{4}{e^2} - \text{b2tqca_taua01}$$

$$_{ay} = \text{ANa} \quad _{CLay}^{(CL)}, \quad _{ay}^{(CL)} = 0 \quad _{ay} \frac{h_z \sqrt{g}}{h_y^2} - \text{b2tral_cvsahzx}$$

$$\tilde{q}^{(0)} = q^{(1)} \arctg \frac{q^{(0)}}{q^{(1)}}, \quad q^{(1)} = c_{q1} n_i T_i \sqrt{\frac{T_i - T_e}{m_i}}$$

$c_{q1} = 0.5$ or `{'b2siav_cqip1' '0.1'}` in b2mn.dat

the three-point smoothing procedure is imposed twice on $\tilde{q}^{(0)}$ along x-direction

in case if `{'b2siav_style_qip' '0'}` in b2mn.dat

$$q^{(0)} = q_{a\parallel}^{(0)} - \frac{B}{B_x} q_{ax}^{(dia)} - \frac{5}{2} \frac{B_z B}{B_x B^2} \frac{n T_i}{e} \frac{T_i}{h_y y} \quad \begin{array}{l} \text{outside the separatrix} \\ \text{inside the separatrix} \end{array}$$

in case if {'b2siav_style_qip' '1'} in b2mn.dat

$$q^{(0)} = q_{a\parallel}^{(0)} - \frac{B}{B_x} q_{ax}^{(dia)} \approx k_{ix}^{(cl)} b_x \frac{T_i}{h_x x} - \frac{5}{2} \frac{B_z B}{B_x B^2} \frac{n T_i}{e} \frac{T_i}{h_y y} \quad \begin{array}{l} \text{outside the separatrix} \\ \text{inside the separatrix} \end{array}$$

plato and banana correction 'b2trcl_lluciani' '3'

$$\sim \frac{a}{15,12 \cdot 10^{16} \frac{T_i}{e}} - \frac{1}{15,12 \cdot 10^{16} \frac{3/2 T_i}{e}} \quad \begin{array}{l} 1 \\ 1 \end{array} \quad \begin{array}{l} c(y) \\ c(y) \end{array} \quad \begin{array}{l} z_a^2 n_a \\ z_a^2 n_a \end{array} \quad \begin{array}{l} a=0 \\ a=0 \end{array}$$

$$NEO \quad \frac{8}{15} \frac{0.17}{1} \frac{1.05}{0.7} \frac{1/2}{1/2} \frac{2.7}{2} \frac{2}{3} \quad \text{gives the correct coefficient before the temperature gradient}$$

in the neoclassical electric field in the plato and banana regime

$$\frac{c y}{2^{3/2} a \sqrt{T_i / m_a}}, \quad c y \quad \circ h_x b_x^{-1} dx, \quad \frac{\circ h_x dx^2}{\circ \sqrt{g} / h_y dx} \quad \begin{array}{l} \frac{3}{2} \frac{n_e T_e}{t} \frac{1}{\sqrt{g}} \frac{1}{x} \frac{\sqrt{g}}{h_x} \tilde{q}_{ex} \quad \frac{1}{\sqrt{g}} \frac{1}{y} \frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \quad \frac{n_e T_e}{\sqrt{g}} \frac{1}{x} \frac{\sqrt{g} b_x}{h_x} V_{e\parallel} \end{array} \quad (3)$$

$$Q_e = c_E B n_e T_e B \frac{1}{h_x h_y} \frac{1}{y} \frac{1}{x} \frac{1}{B^2} \frac{1}{x} \frac{1}{y} \frac{1}{B^2} \quad j_x E_x \quad j_y E_y, \quad \frac{n_e T_e}{\sqrt{g}} \frac{1}{x} \frac{\sqrt{g} b_x}{h_x} V_{e\parallel}$$

T_e - b2nph9_te

The residuals of the iteration of this equation are b2nph9_reshe

$$\frac{\sqrt{g}}{h_{x,y}} \tilde{q}_{ex,y} - \text{b2nph9_fhex,y}$$

$$Q_e = c_E B n_e T_e B \frac{1}{h_x h_y} \frac{1}{y} \frac{1}{x} \frac{1}{B^2} \frac{1}{x} \frac{1}{y} \frac{1}{B^2} \quad j_x E_x \quad j_y E_y \quad \frac{n_e T_e}{\sqrt{g}} \frac{1}{x} \frac{\sqrt{g} b_x}{h_x} V_{e\parallel} \quad \sqrt{g} - \text{b2nph9_she}$$

$$Q_e = Q_{ionization}$$

$Q_{ionization} \sqrt{g} - \text{b2stel_she_rad}$ - includes both ionization and radiation losses

$$Q = c_{eqp} e^{\frac{3}{2}} \frac{n_i}{n_s^{-1} z_a^2 m_p n_a T_e^2} \frac{n_e n_i}{T_e^2} (T_e - T_i), \quad c_{eqp} = 4,8 \cdot 10^{15}, \quad 12, \quad Q \sqrt{g} - \text{b2nph9_shei}$$

$$j_x E_x \quad j_y E_y \quad \sqrt{g} - \text{b2sihs_shefr}$$

$$\frac{n_e T_e}{\sqrt{g}} \frac{1}{x} \frac{\sqrt{g} b_x}{h_x} V_{e\parallel} \quad \sqrt{g} - \text{b2sihs_shedu}$$

$$c_{EB} n_e T_e B \frac{1}{h_x h_y} \frac{1}{y} \frac{1}{x} \frac{1}{B^2} \frac{1}{x} \frac{1}{y} \frac{1}{B^2} \sqrt{g} - \mathbf{b2sihs_shedd}$$

$$\frac{3}{2} \frac{n_e T_e}{t} - \mathbf{b2srdt_shedt}$$

$$\tilde{q}_{ex} \tilde{f}_{ex} T_e \frac{1}{h_x} \frac{T_e}{x} \frac{T_e}{x} E_x q_{ex}^{P.Sch}$$

$$\tilde{f}_{ex} \frac{3}{2} \frac{V_{ex}^{(Str)}}{n_e} \frac{\sqrt{g}}{h_{x,y}} \tilde{f}_{ex,y} T_e - \mathbf{b2tfhe_qe_32x,y}$$

$$q_{ex}^{P.Sch} \frac{5}{2} D_{exy} \frac{1}{h_y} \frac{n_e T_e^2}{y}, \quad \frac{\sqrt{g}}{h_{x,y}} q_{ex,y}^{P.Sch} - \mathbf{b2tfhe_fhePSchx,y}$$

$$\frac{\sqrt{g}}{h_{x,y}} \frac{1}{h_{x,y}} \frac{T_e}{x} \frac{T_e}{y} - \mathbf{b2tfhe_qe_ke_gTx,y}$$

$$T_e E_{x,y} - \mathbf{b2tfhe_qe_alphaTehx,y}$$

$$q_{ex} \frac{3}{2} \frac{5}{2} n_e T_e \frac{B_z}{e h_y y} \frac{1}{B^2} V_{ex}^{(Str)} n_e T_e \frac{1}{h_x} \frac{T_e}{x} T_e E_x; \quad \frac{\sqrt{g}}{h_{x,y}} q_{ex,y} - \mathbf{b2tfhe_fhe_no_mdfx,y}$$

$$n_s \frac{1}{a} \mathbf{z}_a^{(he)} \frac{j_{\parallel} b_x}{e},$$

$$\frac{(he)}{ax} (b_x V_{a\parallel} V^{(E)} V_a^{(AN)} V^{corr-dpc}) n_a \frac{5}{3} D_a^n \frac{1}{h_x} \frac{n_a}{x} \tilde{D}_{ax}^p \frac{1}{h_x} \frac{p_a}{x}$$

$$V_{ex}^{(Str)} = 0$$

$$\frac{(AN)}{e} \approx \frac{(CL)}{ex}, \quad \frac{\sqrt{g}}{h_{x,y}^2} \text{-modified by Potanko correction } \mathbf{b2tfhe_chcex,y_Pat},$$

before the modification $\mathbf{b2tral_chcex,y}$

$$\frac{(AN)}{e} c_{h_e,0} c_{h_e,1} \frac{10^{20}}{n_e} c_{h_e,3} \frac{6,2 \cdot 10^{20} T_e}{e|B|} n_e f_{h_e}(x,y),$$

$$c_{h_e,0} = 1, c_{h_e,1} = 0, c_{h_e,3} = 0,$$

$$f_{h_e}(x,y) = \begin{matrix} 1,h_e, & y=0, & x \text{ Core}, \\ 2,h_e, & 0 < y < y_{2,h_e}, & x \text{ Core}, \\ 3,h_e, & x,y > y_{2,h_e} & \text{Pr SOL} \end{matrix}$$

$$\tilde{c}_{ex}^{(CL)} c_{k_{ex}}^{(Flim)} \sim c_{ex}^{(CL)} \frac{\sqrt{g}}{h_x^2} - \mathbf{b2trcl_luciani_flim_chcex}$$

1, only for inner flux surfaces

$$c_{k_{ex}}^{(F \text{ lim})} = \frac{1}{1 - \frac{\sqrt{g}}{h_x} \sim_{ex}^{(CL)} \left| \frac{T_e}{h_x x} \right|}, \text{ for other flux surfaces}$$

$$V_{h_e}^{(\max)} = \frac{c_{f \text{ lim } 0}}{V_{h_e}^{(\max)}} n_e T_e$$

$$V_{h_e}^{(\max)} = c_{f \text{ lim } 0} \frac{\sqrt{g}}{h_x} b_x \sqrt{\frac{T_e}{m_e}},$$

$$\sim_{ex}^{(CL)} c_{ex}^{(Luciani)} \sim_{ex}^{(CL)} b_x^2, \quad c_{ex}^{(Luciani)} = 1 \quad \sim_{ex}^{(CL)} \frac{\sqrt{g}}{h_x^2} - \mathbf{b2trcl_luciani_chcex}$$

$$\sim_{ex}^{(CL)} \frac{5}{2} \frac{T_e}{m_e} f_{k_e} z_{eff} n_e, \quad \sim_{ex}^{(CL)} \frac{\sqrt{g}}{h_x^2} b_x^2 - \mathbf{b2trcl_chcex}$$

$$f_{k_e} z_{eff} = \frac{2,16 z_{eff}}{1 - 0,27 z_{eff}}, \quad z_{eff} = \frac{z_a^2 n_a}{n_e},$$

$$e = \frac{3}{4\sqrt{2}} \frac{\sqrt{m_e T_e^2}}{z_{eff} n_e} \frac{4}{e^2}^2, - \mathbf{b2tqce_taue}$$

12 ,

$$V_{e\parallel} = \frac{1}{en_e} z_a en_a V_{a\parallel} j_{\parallel},$$

$$j_{\parallel} = E_x \sim_{ex} \frac{1}{h_x} \frac{T_e}{x},$$

$$\parallel \sim_{CLx} b_x, \quad \sim_{CLx} c_{CLx}^{(Luciani)} \sim_{CLx}, \quad \parallel \frac{\sqrt{g}}{h_x^2} b_x - \mathbf{b2trcl_luciani_csigx}$$

$$c_{CLx}^{(Luciani)} = \frac{1}{7,5 \cdot 10^{16} \frac{T_e}{e}},$$

$$1 - \frac{c(y)}{z_a^2 n_a}$$

$$_{CLx} \frac{e^2}{m_e} f_e z_{eff} n_e, \quad _{CLx} \frac{\sqrt{g}}{h_x^2} b_x - \mathbf{b2trcl_csigx}$$

$$f_e z_{eff} = \frac{4,95 z_{eff}}{1 - 1,54 z_{eff}},$$

[illegible]

$$\frac{3}{2} \frac{n_i T_i}{t} - \frac{1}{\sqrt{g}} \frac{1}{x} \frac{\sqrt{g}}{h_x} \tilde{q}_{ix} - \frac{1}{\sqrt{g}} \frac{1}{y} \frac{\sqrt{g}}{h_y} \tilde{q}_{iy} - \frac{n_s - 1}{a - 0} \frac{n_a T_i}{\sqrt{g}} \frac{1}{x} \frac{\sqrt{g}}{h_x} V_{a\parallel} b_x$$

$$Q - Q_{Fab} - c_{EB} T_i B \frac{1}{h_x h_y} - \frac{1}{y} \frac{1}{x} \frac{1}{B^2} - \frac{1}{x} \frac{1}{y} \frac{1}{B^2} - \frac{n_s - 1}{a - 0} \frac{n_a}{z_a - 0}$$

$$\frac{n_s - 1}{a - 0} \frac{4}{3} \frac{V_{a\parallel}^2}{h_x x} - \frac{V_{a\parallel}^2}{h_y y} \quad \text{ANa} \quad ,$$

T_i - **b2nph9_ti**

The residuals of the iteration of this equation are **b2nph9_reshi**

$$\frac{\sqrt{g}}{h_{x,y}} \tilde{q}_{ix,y} - \mathbf{b2nph9_fhix,y}$$

$$\frac{3}{2} \frac{n_i T_i}{t} - \mathbf{b2srdt_shidt}$$

$$(Q - Q_{Fab} - c_{EB} T_i B \frac{1}{h_x h_y} - \frac{1}{y} \frac{1}{x} \frac{1}{B^2} - \frac{1}{x} \frac{1}{y} \frac{1}{B^2} - \frac{n_s - 1}{a - 0} \frac{n_a}{z_a - 0}) - \mathbf{b2nph9_shi}$$

$$\frac{n_s - 1}{a - 0} \frac{4}{3} \frac{V_{a\parallel}^2}{h_x x} - \frac{V_{a\parallel}^2}{h_y y} - \frac{n_s - 1}{a - 0} \frac{n_a T_i}{\sqrt{g}} \frac{1}{x} \frac{\sqrt{g}}{h_x} V_{a\parallel} b_x - \sqrt{g} ,$$

$$\sqrt{g} c_{EB} T_i B \frac{1}{h_x h_y} - \frac{1}{y} \frac{1}{x} \frac{1}{B^2} - \frac{1}{x} \frac{1}{y} \frac{1}{B^2} - \frac{n_s - 1}{a - 0} \frac{n_a}{z_a - 0} - \mathbf{b2sihs_shidd}$$

$$\frac{n_s - 1}{a - 0} \frac{n_a T_i}{x} \frac{\sqrt{g}}{h_x} V_{a\parallel} b_x - \mathbf{b2sihs_shidu}$$

$$\sqrt{g} \frac{n_s - 1}{a - 0} \frac{4}{3} \frac{V_{a\parallel}^2}{h_x x} - \mathbf{b2sihs_shivc}$$

$$\sqrt{g} \frac{n_s - 1}{a - 0} \frac{V_{a\parallel}^2}{h_y y} - \mathbf{b2sihs_shiva}$$

$$Q_{Fab} \sqrt{g} - \mathbf{b2stcx_shi}$$

$$\tilde{q}_{ix} - \tilde{f}_{ix} T_i - \frac{1}{h_x} \frac{T_i}{x} - q_{ix}^{P.Sch} ,$$

$$\tilde{f}_{ix} - \frac{3}{2} \frac{V_{ix}^{(Str)}}{n_i} , \quad \frac{\sqrt{g}}{h_{x,y}} \tilde{f}_{ix,y} T_i - \mathbf{b2tfhi_qi_32x,y}$$

$$q_{ix}^{P.Sch} - \frac{5}{2} \frac{n_s - 1}{a - 0} \frac{D_{axy}}{h_y} \frac{1}{y} \frac{n_a T_i^2}{y} - \frac{\sqrt{g}}{h_{x,y}} q_{ix,y}^{P.Sch} - \mathbf{b2tfhi_fhiPSchx,y}$$

$$_{ix,y}\frac{1}{h_{x,y}}\frac{T_i}{x,y}-b2tfhi_qi_ki_gTx,y$$

$$q_{ix}=\frac{3}{2}{}_{ix}\frac{5}{2}{}_{a\,0}^{n_s-1}\widetilde{n_aV_a^{(dia)}}V_{ix}^{(Str)}n_iT_i{}_{ix}\frac{1}{h_x}\frac{T_i}{x},\qquad V_{ix}^{(Str)}=0$$

$$_{ix}{}_{a\,0}^{n_s-1}\left(he\right)_{ax}$$

$$_{ix}{}_{ix}^{(AN)}\widetilde{{}_{ix}^{(CL)}}b_x^2,$$

$$_{ix,y}\frac{\sqrt{g}}{h_{x,y}^2}\text{-modified by Potanko correction }b2tfhi_chcix,y_Pat,\text{ before the modification }b2tral_chcix,y$$

$$k_{ix}^{(AN)}{}_{a\,0}^{n_s-1}\widetilde{k}_{ax}^{(AN)},$$

$$k_{ax}^{(AN)},\qquad z_a=0$$

$$\widetilde{k}_{ax}^{(AN)}=\frac{k_{ax}^{(AN)}}{1-\frac{\left|\frac{T_i}{h_xx}\right|^2}{V_{h_i}n_aT_i}},\qquad z_a=0,\qquad V_{h_i}=\sqrt{\frac{2T_i}{m_a}}$$

$$k_{ax}^{(AN)}=c_{h_a,0}c_{h_a,1}\frac{10^{20}}{n_e}c_{h_a,2}D_a^fc_{h_a,3}\frac{6,2\cdot10^{-2}T_e}{e|B|}n_af_{h_a}(x,y),$$

$$a=0,\,c_{h_a,0}=0,\,c_{h_a,1}=0,\,c_{h_a,2}=1.4,\,c_{h_a,3}=0,$$

$$a=1,\,c_{h_a,0}=0,\,c_{h_a,1}=0,\,c_{h_a,2}=1.2,\,c_{h_a,3}=0,$$

$$1,\quad z_a=0$$

$$f_{h_a}(x,y)=\begin{matrix}{}_{1,h_a}, & {}_{1,h_a}y=0, & x=Core, & z_a=0 \\ {}_{2,h_a}, & 0 & y=_{2,h_a}x=Core, \\ {}_{3,h_a}, & x,y=Pr & SOL\end{matrix}$$

$$\widetilde{k}_{ix}^{(CL)}=c_{k_{ix}}^{(F\lim)}\widetilde{k}_{ix}^{(CL)}\qquad\qquad\qquad\widetilde{{}_{ix}^{(CL)}}\frac{\sqrt{g}}{h_x^2}-b2trcl_luciani_flim_chcix$$

for inner flux surfaces

$$c = \frac{\sqrt{g}}{h} k_{CL} \frac{T_i}{h_x X} \quad \text{for other flux surfaces},$$

$$V_{h_i}^{(\max)} = c_{f \lim 1} \frac{\sqrt{g}}{h_x} b_x \sqrt{\frac{T_i}{m_p}} \frac{n_a \sqrt{\frac{m_p}{m_a}}}{n_i},$$

$$c_{f \lim 1} = 0,15$$

$$\sim_{ix}^{(CL)} = c_{k_{ix}}^{(Luciani)} \sim_{ix}^{(CL)}, \quad \sim_{ix}^{(CL)} = \frac{\sqrt{g}}{h_x^2} - b2trcl_luciani_chcix$$

For plato and banana correction ‘b2trcl_lluciani’ ‘3’

$$c_{k_{ix}}^{(Luciani)} = 1.6^{3/2} / K_2,$$

$$K_2 = \frac{0.66}{1} \frac{1.88}{1.03} \frac{\sqrt{1.54}}{0.31} \frac{1.17^3}{0.74^{3/2}} *$$

$$\sim_{ix}^{(CL)} = \frac{5^{n_s-1} T_i}{2^{a_0} m_a} n_a f_{k_i a}, \quad f_{k_i} = 2.253, \quad \sim_{ix}^{(CL)} = \frac{\sqrt{g}}{h_x^2} - b2trcl_chcix$$

For integral modeling in the region of the overlap with ASTRA:

$$\tilde{q}_{iy} = \tilde{f}_{iy} T_i \frac{1}{h_y} \frac{T_i}{y} = q_{iy}^{P.Sch} \frac{3}{2} T_i \frac{j_y^{(in)} j_y^{(AN)} j_y^{(vis \parallel)}}{e}$$

$$\tilde{f}_{iy} = \frac{3}{2} \frac{V_{iy}^{(Str)}}{V_{iy}^{(Str)}} n_i, \quad V_{iy}^{(Str)} = 0$$

$$q_{iy}^{P.Sch} = \frac{5^{n_s-1}}{2^{a_0} z_a^0} D_{ayx} \frac{1}{h_x} \frac{n_a T_i^2}{x}$$

$$q_{iy} = \frac{3}{2} \frac{V_{iy}^{(Str)}}{V_{iy}^{(Str)}} n_i \frac{3}{2} \frac{j_y^{(in)}}{e} T_i \frac{1}{h_y} \frac{T_i}{y}$$

0, for us

$S \cot t$ 1, for Bruce $S \cot t$

$$\begin{aligned} & \begin{matrix} n_s - 1 \\ i_y \end{matrix} \begin{matrix} (he) \\ a = 0 \end{matrix} \\ & \begin{matrix} (AN) \\ i_y \end{matrix} \begin{matrix} (CL) \\ i_y \end{matrix}, \quad \begin{matrix} (CL) \\ i_y \end{matrix} = 0 \end{aligned}$$

$$\begin{aligned} k_{i_y}^{(AN)} &= \begin{matrix} n_s - 1 \\ a = 0 \end{matrix} \tilde{k}_{ay}^{(AN)}, \\ k_{ay}^{(AN)}, & \quad z_a = 0 \\ & \frac{k_{ay}^{(AN)}}{\frac{1}{2}}, \quad z_a = 0 \\ \tilde{k}_{ay}^{(AN)} &= \frac{1}{V_{h_i} n_a T_i} \frac{\sqrt{g} k_a^{(AN)} \left| \frac{T_i}{h_y y} \right|}{1} \end{aligned}$$

$$\frac{1}{\sqrt{g}} \frac{\sqrt{g}}{h_x} j_x = \frac{1}{\sqrt{g}} \frac{\sqrt{g}}{h_y} j_y = 0$$

- **b2npp7_po**

The residuals of the iteration of this equation are **b2npp7_respo**

$$\begin{aligned} & \frac{\sqrt{g}}{h_{x,y}} j_{x,y} - \mathbf{b2nph9_fhix,y} \\ j_x &= j_x^{(AN)} = \tilde{j}_x^{(dia)} = j_x^{(in)} = \tilde{j}_x^{(vis \parallel)} = \tilde{j}_x^{(vis)} = \tilde{j}_x^{(visq)} = \tilde{j}_x^{(s)} = j_{\parallel}, \end{aligned}$$

$$j_{\parallel} \text{ -is not multiplied by } b_x \text{ because of } \sigma_{CLx} \text{ and } \alpha_{ex} \text{ is multiplied by } b_x, \quad \frac{\sqrt{g}}{h_x} j_{\parallel} - \mathbf{b2tfhe_fch_px}$$

$$\begin{aligned} \tilde{j}_x^{(dia)} &= B_z \frac{n_a z_a T_e T_i}{h_y} \frac{1}{y B^2}, \quad \frac{\sqrt{g}}{h_{x,y}} \tilde{j}_{x,y}^{(dia)} - \mathbf{b2tfhe_fchdia_{x,y}} \\ j_x^{(in)} &= \frac{B_z}{2 h_y y} \frac{1}{B^2} m_a n_a V_{a\parallel}^2, \quad \frac{\sqrt{g}}{h_{x,y}} j_{x,y}^{(in)} - \mathbf{b2tfhe_fchinert_{x,y}} \\ \tilde{j}_x^{(vis \parallel)} &= \frac{b_z B_x}{3 B^2} \frac{1}{h_y y} \frac{1}{B^2} \frac{n_s - 1}{z_a = 0} \frac{\sqrt{B} V_{a\parallel}}{h_x x}, \quad \frac{\sqrt{g}}{h_{x,y}} \tilde{j}_{x,y}^{(vis \parallel)} - \mathbf{b2tfhe_fchvispar_{x,y}} \\ \tilde{j}_x^{(vis)} &= 0, \quad \tilde{j}_x^{(visq)} = 0, \end{aligned}$$

$$\tilde{j}_x^{(s)} = a_0 b_z^2 \frac{1}{h_x x} = a_0 b_z^2 \frac{1}{Z_a e n_a} \frac{n_a T_i}{h_x x} = a_0 B_z V_{y0}, \quad a = 1, \quad \frac{\sqrt{g}}{h_{x,y}} \tilde{j}_{x,y}^{(s)} - \mathbf{b2tfhe_fchin_{x,y}}$$

$$_{a\,0}\frac{m_a n_a \left\langle V_{a\,0\,ex} \right\rangle n_0}{2\,B^2},\; a=1,$$

$$_{a\,0}b_z^2\frac{\sqrt{g}}{h_{x,y}^2}-b2tral_csiginx01_00$$

$$\left\langle V_{a\,0\,ex} \right\rangle=3.2\cdot 10^{-15}\sqrt{\frac{T_i}{0.026\,e}}$$

$$V_{y0}=\frac{_{0\,y}}{n_0}$$

$j_{\parallel}$ - see above

$$j_y=j^{(AN)}\tilde{j}_y^{(dia)}j_y^{(in)}\tilde{j}_y^{(vis\parallel)}j_y^{(vis)}\tilde{j}_y^{(visq)}\tilde{j}_y^{(s)}\tilde{j}_y^{(stoch)},$$

$$j^{(AN)}_{\text{ANy}}\frac{\hspace{1cm}}{h_y\hspace{0.1cm}y},\hspace{0.5cm}\frac{\sqrt{g}}{h_{x,y}}\tilde{j}_{x,y}^{(AN)}-b2tfhe_csig_anx,y$$

$$\tilde{j}_y^{(dia)}=\frac{B_z\frac{n_a\,z_aT_e}{\frac{a\,0}{z_a\,0}}T_i}{h_x}\frac{1}{x\,B^2},$$

$$j_y^{(in)}=\frac{B_z}{2\,h_x\,x}\frac{1}{B^2}\frac{n_s-1}{\frac{a\,0}{z_a\,0}}m_a n_a V_{a\parallel}^2,$$

$$\tilde{j}_y^{(vis\parallel)}=b_z\frac{B_x}{3B^2}\frac{1}{\frac{1}{h_x\,x}}\frac{1}{B^2}\frac{n_s-1}{\frac{a\,0}{z_a\,0}}\frac{B^{\frac{1}{2}}V_{a\parallel}}{h_x\,x},$$

$$j_y^{(vis)}=\frac{1}{B\sqrt{g}}\frac{1}{y}\frac{\sqrt{g}}{h_y}\frac{ANa}{4}\frac{V_a}{h_y\,y},\; a=1,\hspace{1cm}\frac{\sqrt{g}}{h_y}j_y^{(vis)}-b2tfhe_fchvispery$$

$$V_a=\frac{1}{Bh_y\,y}-\frac{1}{z_a\,eBh_y\,n_a}\frac{n_a\,T_i}{y}$$

$$\tilde{j}_y^{(visq)}=b_z\frac{B_x}{3B^2}\frac{1}{\frac{1}{h_x\,x}}\frac{1}{B^2}\overset{NEO}{\sim}_a\frac{B^{\frac{1}{2}}\tilde{q}^{(0)}}{h_x\,x},\; a=1,$$

$$\tilde{j}_y^{(s)}=\frac{_{a0}}{h_y\,y}=\frac{1}{_{a0}\frac{Z_aen_a}{h_y\,y}}\frac{n_aT_i}{h_y\,y}=\frac{_{a0}B\,V_0}{_{a0}},\; a=1,\hspace{1cm}V_0=\frac{_{0\,x}}{n_0}$$

$$_y\text{ANy}\text{CLy},\hspace{1cm}\text{CLy}\hspace{0.5cm}0$$

Any $c_{a,0} \quad c_{a,1} \frac{10^{20}}{n_e} \quad c_{a,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} \quad c_{a,4} e^{c_{a,5} \frac{y}{Lx}^2} e\tilde{n}_e, \tilde{n}_e$ density at inner flux surface

$L \cdot x \int_0^{y_{out}} h_y(x,y) dy \quad c_{a,0} \cdot 10^{-6} \cdot 10^{-3}, c_{a,1} = 0, c_{a,3} = 0$

EIRENE output

b2stbr_sna_eir01 – ionisation source of ions from EIRENE

b2stbr_smo_eir01 = b2stbr_smo_eir001 + b2nph9_ua01*b2stbr_smo_eir101 source of parallel momentum for 01 sort of ions

b2stbr_she_eir = b2stbr_she_eir0 + b2nph9_te*b2stbr_she_eir1 source of energy for electrons

b2stbr_shi_eir = b2stbr_shi_eir0 + b2nph9_ti*b2stbr_shi_eir1 source of energy for ions

b2stbr_dab2_eir01 – atom density for 01 sort of atoms

b2stbr_dmb2_eir01 – atom density for 01 sort of molecules

b2stbr_pfluxa_eir01– poloidal flux (particles to meter² to second) for 01 sort of atoms

b2stbr_rfluxa_eir01– radial flux (particles to meter² to second) for 01 sort of atoms

b2stbr_pfluxm_eir01– poloidal flux (particles to meter² to second) for 01 sort of molecules

b2stbr_rfluxm_eir01– radial flux (particles to meter² to second) for 01 sort of molecules